

URL Input Function

The URL input function allows users to paste or type a suspicious link into PhishGuard for review. This is the starting point of the prototype because the user must provide a URL before the system can perform any analysis. The purpose of this function is to make the tool simple and direct, allowing users to quickly check a link before deciding whether it is safe to open.

URL Scan Function

The URL scan function reviews the submitted link for possible phishing indicators. The scan looks for warning signs such as missing HTTPS, suspicious keywords, excessive URL length, URL shorteners, and possible redirect behavior. This function supports the main purpose of PhishGuard by helping users identify whether a link may contain suspicious characteristics.

Risk Score Function

The risk score function gives the user a clear risk level based on the warning signs found during the scan. Instead of only showing technical details, this function helps summarize the result in a way users can quickly understand. The score can help users decide whether they should avoid the link, investigate further, or treat it with caution.

Sonar Analysis Log

The Sonar Analysis Log displays the specific warnings detected during the scan. This section helps users see what caused the risk score instead of only receiving a final result. For example, the log may show issues such as “Missing HTTPS,” “URL Shortener Detected,” or “Suspicious Keyword Found.” This supports PhishGuard’s goal of explainable phishing risk analysis.

Bait Breakdown Feature

The Bait Breakdown feature expands the Sonar Analysis Log by explaining each warning in simple language. For each detected issue, the feature would explain what was found, why it may be risky, and what the user should do next. This feature supports cybersecurity awareness because it helps users learn from the scan instead of only seeing a warning label.

Recommended Action Function

The recommended action function gives users a practical next step after the scan is complete. Depending on the result, PhishGuard may advise the user not to click the link, verify the sender, avoid entering personal information, or visit the official website directly. This function supports safer digital decision making by helping users respond appropriately to suspicious content.

Cybersecurity Awareness Guidance

The cybersecurity awareness guidance function provides short educational notes connected to the scan results. These notes help users understand common phishing tactics, such as fake login pages, urgent language, suspicious links, and impersonation. This function supports PhishGuard’s educational goal by helping users recognize similar risks in the future.

Result Display Function

The result display function presents the scan outcome in a clear and organized way. It should include the risk score, the warnings found, the explanation of those warnings, and the recommended action. This function is important because it brings all parts of the analysis together and allows the user to understand the final result without needing technical knowledge.

Documentation and Artifact Support

This function is not part of the user interface, but it supports the project development process. It connects the prototype features to the project documentation, research sources, user stories, acceptance criteria, and risk assessment. This helps ensure that the prototype is aligned with the overall Capstone 1 project scope.